Journal of Vibration Engineering & Technologies (2019) 7:465–475
https://doi.org/10.1007/s42417-019-00144-x

ORIGINAL PAPER

Research on Energy-Regenerative Performance of Suspension System with Semi-active Control

D. H. Shi[1] · L. Chen[1] · R. C. Wang[1] · C. C. Yuan[1] · Y. L. Liu[2]

Received: 5 September 2017 / Accepted: 20 April 2018 / Published online: 12 June 2019
© Krishtel eMaging Solutions Private Limited 2019

Abstract

Background If the vibration energy of traditional vehicle shock absorber is harvested and reused, the vehicle fuel consumption and exhaust emission will be further reduced. Various transducers of energy-regenerative suspension have been proposed. Semi-active control is applied to coordinate the energy-regenerative performance and the vehicle vibration isolation performance. To realize semi-active control, it is important to design appropriate strategies and energy-regenerative circuit for practical use.

Method The unidirectional boost–buck converter is introduced. The steady characteristics of the converter are revealed through the establishment of dynamic differential equations in different operation modes, and thus, the theoretical basis for the mode switch strategy of the converter is provided. Then, the direct current controller including a mode decision module and a sliding mode controller is designed, where the sliding mode strategy is designed based on the average models of the converter under continuous conduction mode. The semi-active control force is derived according to sky-hook and ground-hook control. The relations between semi-active control force and the maximum electromagnetic damping force in frequency domain is theoretically analyzed to increase the effective bandwidth of controllable semi-active force.

Results and conclusions Simulation results demonstrate that the energy-regenerative suspension with proposed circuit and semi-active control strategy improves the vehicle ride comfort obviously on random excitation road while slightly deteriorates road-holding performance. The energy-regenerative suspension ensures a 20.53 W of energy that charges the battery on D-class road when the vehicle velocity is 60 km/h.

Keywords Energy-regenerative suspension · Semi-active control · Converter · Sliding mode control · Performance coordination

Introduction

Traditionally, the vibration energy of the vehicle suspension system is dissipated by the oil shock absorber in the form of heat energy to ensure vehicle handling stability and ride comfort. If the dissipated energy is harvested and reused, the vehicle fuel consumption and exhaust emission will be further reduced. Abundant studies on the recovery techniques of suspension vibration energy have been carried out. The feasibility of vibration energy recovery has been validated through both theoretical and experimental analysis [1, 2]. Various transducers of energy-regenerative suspension have also been proposed, both rotational and linear [3].

The energy-regenerative suspension combined with semi-active or active control has been studied to better exert their superior performance. Suda et al. proposed the suspension system that combined energy regeneration and active control for better ride comfort [4]. Okada et al. introduced passive and active control strategies at low-speed motion [5]. And the energy is regenerated in the high-speed scenario. Okada and Kim et al. later introduced a PWM step-up chopper, which improved the damping characteristics and energy-regeneration efficiency [6]. By calculating the balance between the consumed and regenerated energies, Nakano et al. proposed self-powered active vibration control system [7]. Such control system is also applied to the truck cabins suspensions [8]. Montazeri-Gh studied the

✉ D. H. Shi
landy989@126.com

1 Automotive Engineering Research Institute, Jiangsu University, Zhenjiang 212013, People's Republic of China

2 School of Automobile and Traffic Engineering, Jiangsu University, Zhenjiang 212013, People's Republic of China

active control strategy of electromagnetic suspension system, and the control parameters were optimized [9]. As for the semi-active control of energy-regenerative suspension, it is important to design the system for practical use, i.e., the energy-regenerative circuit of electromagnetic transducers. Nakano et al. designed electric circuits that were used to improve the suspension performance in combination with corresponding control strategies [7, 8]. Chen et al. developed a switched-mode rectifier to realize the trade-off between energy consumption and ride handling/comfort with sky-hook strategy [10].

In the paper, a unidirectional boost–buck converter is introduced. By controlling the converter and choosing proper semi-active control parameters, the suspension is controlled without consuming external energy. The overall scheme and semi-active control principle are introduced in Sect. 2. Section 3 discusses the working characteristics of the converter. The direct current controller is designed in Sect. 4. Section 5 theoretically analyzes the impact of control parameters on the effective bandwidth to achieve semi-active control. The control parameters are further decided. Section 6 illustrates the simulation results of the system. Finally, the conclusions are presented in Sect. 7.

Scheme of the System

The System Semi-active Control Principle

Usually, the linear motor is controlled actively to achieve good vibration isolation performance [11, 12]. However, active control usually consumes large amounts of energy [13]. In the paper, semi-active control of the energy-regenerative suspension is studied. Figure 1 demonstrates the topology of the proposed system, where the traditional oil shock absorber is replaced by a linear motor. In the system, the linear motor operates as a generator or electromagnetic damper to convert vibration energy into electrical energy. The electrical energy is stored into the energy storage element (battery) through energy-regenerative circuit. The current controller provides commands to control the circuit.

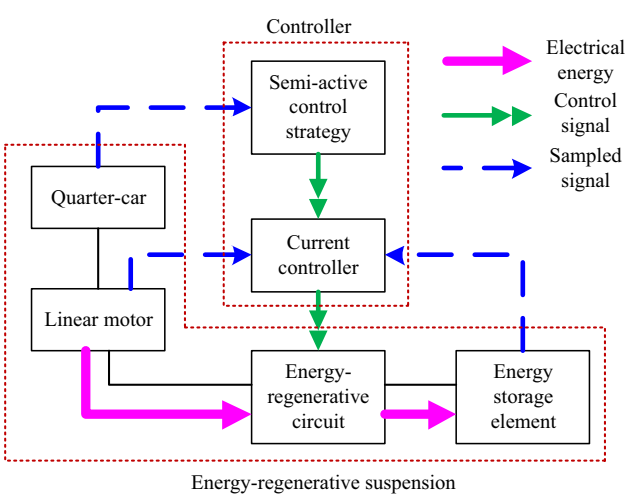


Fig. 1 Topology of the energy-regenerative suspension system

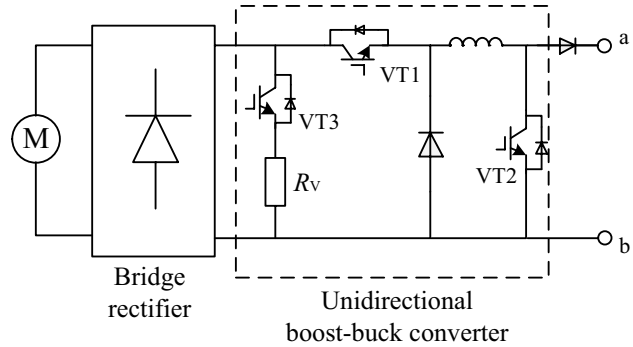


Fig. 2 Topology scheme of energy-regenerative circuit

Energy-Regenerative Circuit

The topology scheme of the circuit is shown in Fig. 2, which contains a diode bridge rectifier and a unidirectional boost–buck converter. The output side (port a, b) can be connected to the battery or ultra capacitor. When VT1 is on and VT2 is provided with PWM signal, the converter operates in boost mode. When VT2 is on and VT1 is provided with PWM signal, the converter is in buck mode. Control signal of VT3 is contrary to that of VT1.

Mathematical Model and Analysis

Suspension Model

The linear motor is applied in the suspension to realize semi-active control and energy regeneration. Figure 3 describes the 2-DOF model, where m_2 is the sprung mass,

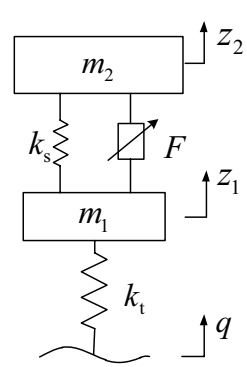


Fig. 3 2-DOF model of the suspension

m_1 is the unsprung mass, z_2 and z_1 are the displacement of sprung mass and unsprung mass, k_s and k_t are the suspension stiffness and tire stiffness, respectively, q is the road displacement.

The differential equations that represent the dynamic behavior of suspension can be expressed as

$$m_2\ddot{z}_2 = k_s(z_1 - z_2) + F, \tag{1}$$

$$m_1\ddot{z}_1 = -k_s(z_1 - z_2) - k_t(z_1 - q) - F, \tag{2}$$

where F is the motor electromagnetic damping force, which is related to the motor winding current and is given by

$$F = k_i i, \tag{3}$$

where k_i and i are the motor thrust constant and winding current, respectively.

When the motor just operates as a generator without being powered by external source, it is essentially an electromagnetic damper. Then, the winding current has the maximum value i_{max} when VT1 and VT2 both switch on and the windings are directly connected.

$$E = k_e v, \tag{4}$$

$$i_{max} = \frac{E}{R}, \tag{5}$$

where E is the motor back electromotive force (EMF), R denotes the internal resistance of the motor windings, k_e represents the linear motor back EMF constant, v is the motor working velocity, i.e., $v = z_2 - z_1$.

According to (3–5), the maximum damping force F_{max} can be derived when the windings are connected.

$$F_{max} = \frac{k_i k_e}{R} v, \tag{6}$$

where $c_e = k_i k_e / R$ can be defined as the motor damping coefficient. When the required control force F is smaller than F_{max} and in the same direction as F_{max}, it can be obtained by controlling the energy-regenerative circuit. To achieve semi-active control with continuously adjustable electromagnetic force, F should meet the following condition.

$$\alpha = \left|\frac{F}{F_{max}}\right| \le 1. \tag{7}$$

According to (3), (7) can also be written by

$$\alpha = \left|\frac{i}{i_{max}}\right| \le 1,$$

where α is defined as the tracking constant, which describes the ratio of the motor current to its potential maximum current when the motor just generates electricity.

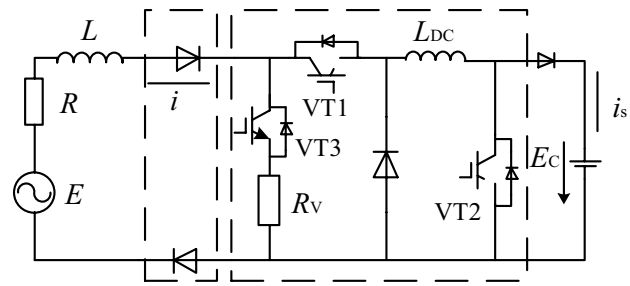


Fig. 4 Circuit model of suspension

Equivalent Circuit Model of Suspension

Figure 4 describes the system equivalent circuit model. In the figure, i_s is the current flows through the battery, L_{DC} is the converter inductance, R_V is the resistance of the converter, E_C is the terminal voltage of the battery and R is the winding inductance.

Supposing that the MOSFET switching period is T, and the switch-on time is t_{on}. Then the duty cycle D ($0 \leqq D \leqq 1$) equals t_{on}/T. It is worth noting that the switching period T is usually very small, the back EMF E is viewed as unchanged.

Boost Mode

VT2 is on ($0 < t \le t_{on}$).

The battery is open-circuited. Based on Kirchhoff Voltage Law (KVL), we have

$$(L + L_{DC})\frac{di}{dt} + iR = E. \tag{8}$$

If the initial winding current is I_{10} at this stage, the winding transient current is derived from (8) as

$$i_1 = I_{10}e^{-\frac{t}{\tau}} + \frac{E}{R}\left(1 - e^{-\frac{t}{\tau}}\right), \tag{9}$$

where $\tau = (L + L_{DC})/R$.

VT2 is off ($t_{on} < t \le T$).

The linear motor directly charges the battery, and the circuit model is

$$(L + L_{DC})\frac{di}{dt} + iR + E_C = E. \tag{10}$$

At this stage, assuming that the converter operates in continuous conduction mode (CCM) [14], the initial current value I_{20} equals the end current value of the previous stage, i.e.

$$I_{20} = i_1(t_{on}). \tag{11}$$

When VT2 is off, the winding current equals the current flowing through battery. We have

$$i_2 = I_{20}e^{-\frac{t-t_{on}}{\tau}} + \frac{E-E_C}{R}\left(1-e^{-\frac{t-t_{on}}{\tau}}\right). \quad (12)$$

Also, I_{10} can be considered to meet the following equation

$$I_{10}=i_2(T). \quad (13)$$

From (9) and (11–13), we have

$$I_{10} = \frac{E}{R} - \frac{E_C}{R}\left(\frac{1-e^{-\frac{T-t_{on}}{\tau}}}{1-e^{-\frac{T}{\tau}}}\right), \quad (14)$$

$$I_{20} = \frac{E}{R} - \frac{E_C}{R}\left(\frac{e^{-\frac{t_{on}}{\tau}}-e^{-\frac{T}{\tau}}}{1-e^{-\frac{T}{\tau}}}\right). \quad (15)$$

If the inductance of L is infinite, (14) and (15) could be simplified with Taylor series and the following equation is obtained

$$I=I_{10}=I_{20}=\frac{E-(1-D)E_C}{R}, \quad (16)$$

where I is assumed as the motor steady winding current under the following conditions: the converter is in boost mode; the duty cycle of VT2 is D; the linear motor back EMF and the battery terminal voltage are E and E_C, respectively.

Buck Mode

The circuit mathematical model in buck mode can also be derived according to the KVL. When VT1 is on ($0<t\le t_{on}$)

$$\left(L+L_{DC}\right)\frac{di}{dt} + iR + E_C = E. \quad (17)$$

During this phase, the winding current and the current flowing through the battery are the same, that is $i_1=i_{s1}$.

$$i_1 = I_{10}e^{-\frac{t}{\tau}} + \frac{E-E_C}{R}\left(1-e^{-\frac{t}{\tau}}\right). \quad (18)$$

When VT1 is off ($t_{on}<t\le T$), the current flowing through the battery will not be reduced to zero immediately. It still flows through the diode, and the electrical energy in the inductance of the previous stage is stored in the battery.

$$L\frac{di}{dt} + i\left(R+R_V\right) = E. \quad (19)$$

Assuming I_{20} is the initial value, then i_2 is given by

$$i_2 = I_{20}e^{-\frac{t-t_{on}}{\tau'}} + \frac{E}{R+R_V}\left(1-e^{-\frac{t-t_{on}}{\tau'}}\right), \quad (20)$$

where $\tau' = L/\left(R+R_V\right)$.

When L is infinite, $L\approx L+L_{DC}$. By assuming $\tau=L/R_V$, we can derive

Table 1 Variation range of winding current

Operating modes	Variation range of winding current $i/(A)$	
	$E\le E_C$	$E>E_C$
Boost	$0\le i<E/R$	$(E-E_C)/R\le i<E/R$
Buck	0	$E/(R+R_V)\le i<(E-E_C)/R$
None	0	$(E-E_C)/R$

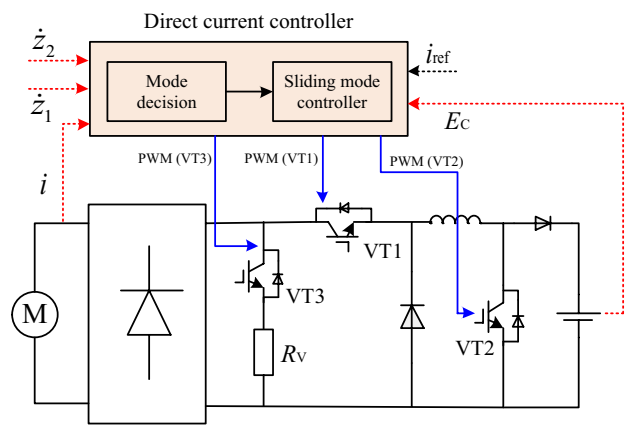


Fig. 5 Scheme of current controller

$$\frac{1}{\tau'} = \frac{1}{\tau} + \frac{1}{\tau_V}. \quad (21)$$

According to (17–21), the motor steady winding current in buck mode is given by

$$I=I_{10}=I_{20}=\frac{E-DE_C}{\left(R+R_V\right)-DR_V}. \quad (22)$$

The steady winding current equals 0 when R_V is infinite and $D=0$. This is the case that motor windings are disconnected.

Current Tracking

Based on (16) and (22), the variation range of winding current with different E and E_C is obtained and shown in Table 1. The winding current and the motor force equals zero when $E\le E_C$ without the proposed converter. When the converter is in boost mode, the current range ($E/(R+R_V)$, $(E-E_C)/R$) cannot be achieved if $E>E_C$. While the converter is in buck mode, if $E\le E_C$, the winding current is always zero, and the motor force is zero. This is obviously bad for the suspension performance.

Aiming at the proposed converter, the direct current controller is designed to realize the current tracking. When the

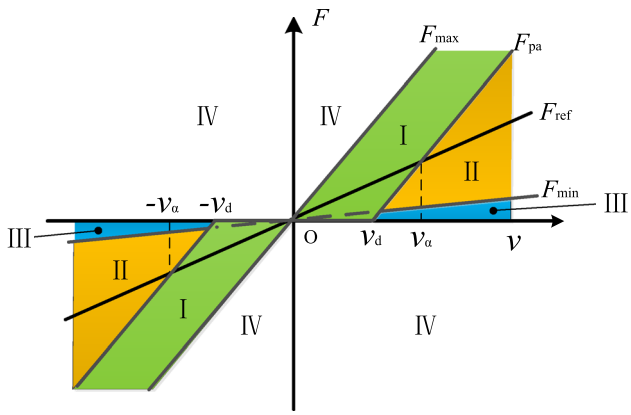


Fig. 6 Motor force in different operation modes

reference force is in the opposite direction of F_{max}, F_{ref} is set to be a small value. The controller scheme is shown in Fig. 5. In the figure, the mode decision module is used to judge the converter operation mode. Then, the duty cycles of different MOSFET are obtained through the sliding mode controller. The motor EMF is calculated according to the suspension velocity. The feedback signals also include the motor winding current and the battery voltage.

Mode Decision

The relations between motor force and velocity are described in Fig. 6. In the figure, the motor is supposed to operate in energy consumption mode when the direction of motor force and velocity are different. Feasible regions of the motor force in different operation modes of the converter can be obtained according to Table 1. In Fig. 6, $F_{max}-v$ and $F_{min}-v$ define the operation boundary when linear motor acts as a generator or electromagnetic damper. $F_{max}-v$ curve is the maximum electromagnetic damping force the motor could generate when the windings are directly connected, while $F_{max}-v$ curve is the minimum damping force when the motor windings are connected via a external resistor R_v. $F_{pa}-v$ curve is obtained when the battery is directly charged by the motor. $F_{ref}-v$ represents the required output of the motor.

The four curves and v-axis divided the $F-v$ plane into regions I, II, III and IV. When the linear motor operates in region IV, the motor needs to be powered by the battery and operates in energy consumption mode, while in region I, II and III, the motor can be controlled without being powered by the battery and recover vibration energy. However, region III is not accessible by controlling the proposed converter. In fact, when E_C and the resistance of external resistor R_V are proper, the linear motor seldom operates in region III. When the converter is in boost and buck mode, linear motor can operate in region I and II, respectively. According to the distribution of motor operation points when the converter operates in different modes, the mode decision law is designed.

It is remarkable that there are two vital working velocities of the linear motor should be highlighted. One is the critical velocity v_d to eliminate 'dead zone' when the battery is directly charged by the linear motor. It also decides the width of motor operation region when the converter operates in boost mode. v_d could be expressed as

$$v_d = \frac{E_C}{k_e}. \tag{23}$$

The other one is v_α which determines the converter operating mode. It can be computed from the following equation

$$\alpha\frac{k_e v_\alpha}{R} = \frac{k_e v_\alpha - E_C}{R}. \tag{24}$$

When $v < v_\alpha$, the converter should operate in boost mode to increase the motor current, and when $v > v_\alpha$, the converter should operate in buck mode to decrease the motor current.

Sliding Mode Controller

PI controller cannot ensure good dynamic performance with the variation of circuit parameters. Sliding mode control turns out to be an effective method for the convert nonlinear characteristics and similar structures [15–17]. In the study, sliding mode control is adopted to maintain ideal current tracking. The average models of the converter under CCM are derived as

$$\dot{i}_L = \begin{cases} -\dfrac{R}{L+L_{DC}} i_L - \dfrac{1-D}{L+L_{DC}} E_C + \dfrac{1}{L+L_{DC}} E & boost \\ -\dfrac{DRL + (1-D)(R+R_V)(L+L_{DC})}{L(L+L_{DC})} i_L - \dfrac{D}{L+L_{DC}} E_C + \dfrac{DL + (1-D)(L+L_{DC})}{L(L+L_{DC})} E & buck. \end{cases} \tag{25}$$

To eliminate the steady-state error. The sliding line is written as

$$s = (i_{ref} - i) + c\int_0^t (i_{ref}(\tau) - i(\tau))\mathrm{d}\tau \quad (c > 0). \tag{26}$$

c is denoted by c_{bo} for boost mode and c_{bu} for buck mode. The equivalent switching control is adopted in the sliding mode controller and is described by

$$u = u_{eq} + u_{sw}, \tag{27}$$

where u_{eq} and u_{sw} are the equivalent control term and switching control term, respectively. Since the control variable is the duty cycle D in each mode, u is limited to be $0 \le u \le 1$. The switching control law of sliding mode control is

$$u_{sw} = k\ \mathrm{sgn}(s), \tag{28}$$

where k is the switching gain. It is denoted by k_{bo} and k_{bu} for boost mode and buck mode, respectively. The equivalent control u_{eq} can be obtained when the system reaches sliding mode (i.e., $\mathrm{d}s/\mathrm{d}t = 0$).

$$u_{eq} = \begin{cases} \dfrac{E_C - E + Ri_{ref} + c_{bo}e\left(L + L_{DC}\right)}{E_C} & \text{boost} \\ \dfrac{E - \left(R + R_V\right)i_{ref} - c_{bu}eL}{\frac{LRi_{ref}}{L+L_{DC}} - \left(R + R_V\right)i_{ref} + \frac{(E_C - E)L}{L+L_{DC}} + E} & \text{buck.} \end{cases} \tag{29}$$

Effective Bandwidth of Semi-active Control

Aiming at the suspension semi-active control, various algorithms have been proposed, such as skyhook control, acceleration driven damp control, ground-hook control, MPC control, etc. [18–20]. Among different strategies, skyhook and ground-hook control are widely adopted. Quantitative analysis of their impacts on the suspension performance could also be realized. Considering that skyhook control is ride comfort-oriented and ground-hook control is road-holding oriented, the reference control force F_{ref} is described as [7]

$$F_{ref} = -c_s\dot{z}_2 - c_g\dot{z}_1, \tag{30}$$

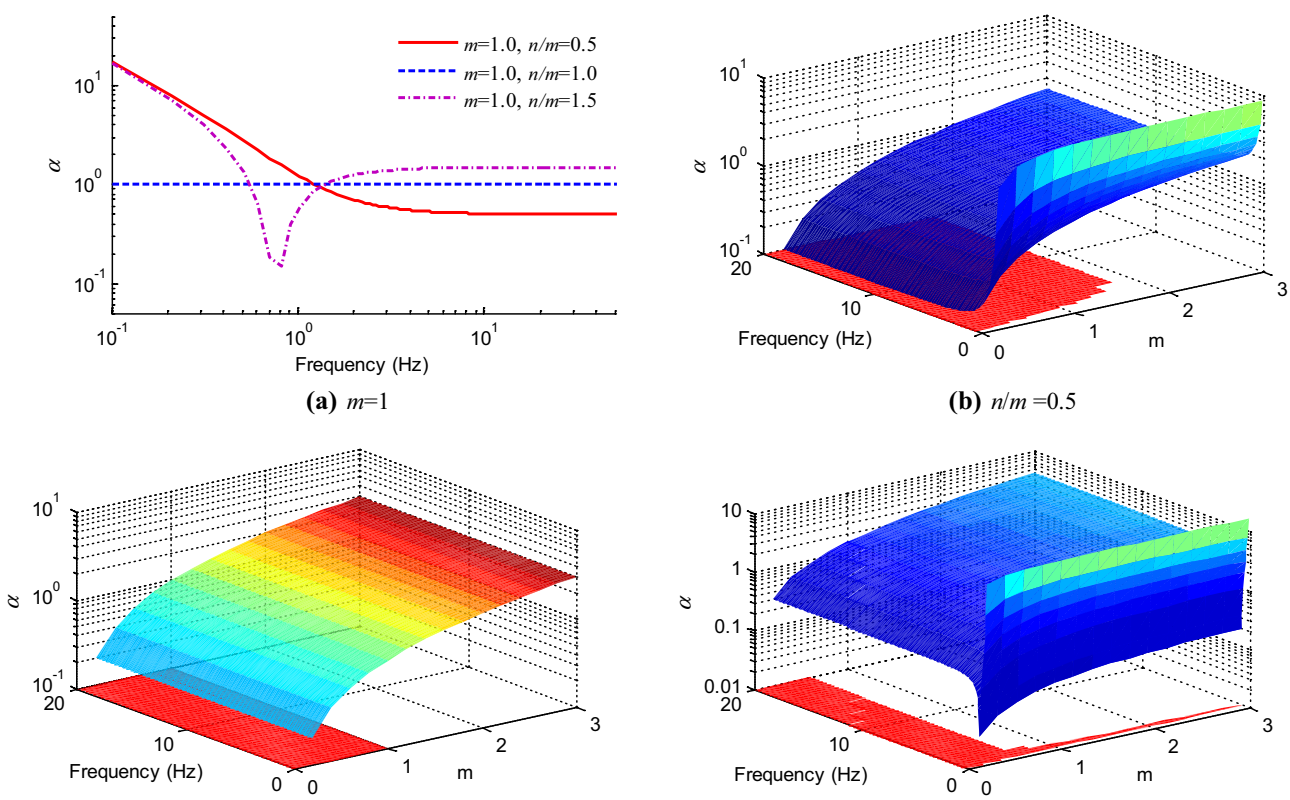
where c_s and c_g are control gains.

The relations between F_{ref} and F_{max} in the frequency domain are derived through Laplace transform.

$$\alpha = \left|\frac{F_{ref}(s)}{F_{max}(s)}\right| = \left|\frac{-c_g m_2 s^2 - k_s\left(c_s + c_g\right)}{c_e m_2 s^2 + c_e\left(c_s + c_g\right)s}\right|. \tag{31}$$

To ensure the stability of the energy-regenerative suspension, $c_s > 0$ and $c_g \le 0$ are necessary [21]. c_s and c_g can be described by $c_s = mc_e$ and $c_g = -nc_s$, respectively, where m and n is bigger than 0. (30) can be expressed as

$$\alpha = \left|\frac{nm_2(i2\pi f)^2 - k_s(m - n)}{m_2(i2\pi f)^2 + c_e(m - n)(i2\pi f)}\right|. \tag{32}$$



(a) m=1 **(b)** n/m =0.5 **(c)** n/m =1 **(d)** n/m =1.5

Fig. 7 Surface of α in frequency domain

When $\alpha < 1$, the linear motor can provide enough electromagnetic damping force as a generator, and the desired F_{ref} is available by controlling the unidirectional boost–buck converter. Therefore, semi-active control of energy-regenerative suspension is realized. However, when $\alpha > 1$, external source, such as battery and ultra capacitor, is required to power the linear motor. While $\alpha = 1$, the linear motor equals a passive electromagnetic damper with the damping coefficient being c_e. Considering the force limitations of the actuator, scale model parameters of a commercial sedan are chosen. The sprung mass and unsprung mass are 82 kg and 11.5 kg. The suspension stiffness is 5495 N/m, and the tire stiffness is 48000 N/m. The passive damping coefficient is 370 N s/m.

It can be found that the curve of α has different shapes when $n/m > 1$, $n/m = 1$ and $n/m < 1$ with the same m. To show the different curve shapes, $n/m = 1.5$, 1 and 0.5 are selected, as shown in Fig. 7a, where m equals 1. Figure 7b–d denote the responses of α with the change of m when $n/m = 0.5$, 1 and 1.5, respectively. The projections on the Frequency-m plane in Fig. 7b–d represent the regions where $\alpha < 1$. The following preliminary conclusions can be obtained:

- When $n/m = 1$, $\alpha = m$ is derived. Therefore, when $m < 1$, $\alpha < 1$ can be realized at all frequencies, as shown in Fig. 7a, c.
- When $n/m < 1$, $\alpha < 1$ cannot be realized at very low frequencies. As for other frequencies, when m is not large, α is smaller than 1, as shown in Fig. 7a, b.
- When $n/m > 1$, there is a through around 1 Hz where $\alpha < 1$. And with a small m, $\alpha < 1$ can be realized at nearly all frequencies, except at very low frequencies. While m becomes large, $\alpha < 1$ is only obtained around 1 Hz, as shown in Fig. 7a, d.
- To make α smaller than 1, m should not be large.
- $n/m < 1$ is a better choice than $n/m > 1$ for that it provides a wider available region of m to realize $\alpha < 1$. Therefore, it is also helpful to choose proper m for suspension vibration isolation. In fact, it is found that a better vibration isolation performance of suspension is always accompanied by $n/m < 1$.

The same m and n cannot maintain good ride comfort and road-holding performance. When n/m is not equal to 1, $\alpha < 1$ cannot be realized at low frequencies. In this case, both VT1 and VT2 are controlled to switch on and the linear motor is controlled to act as a passive electromagnetic damper. Considering that the battery is open-circuited and the damping characteristics is nonadjustable in this case, the frequency region where $\alpha > 1$ should be narrowed.

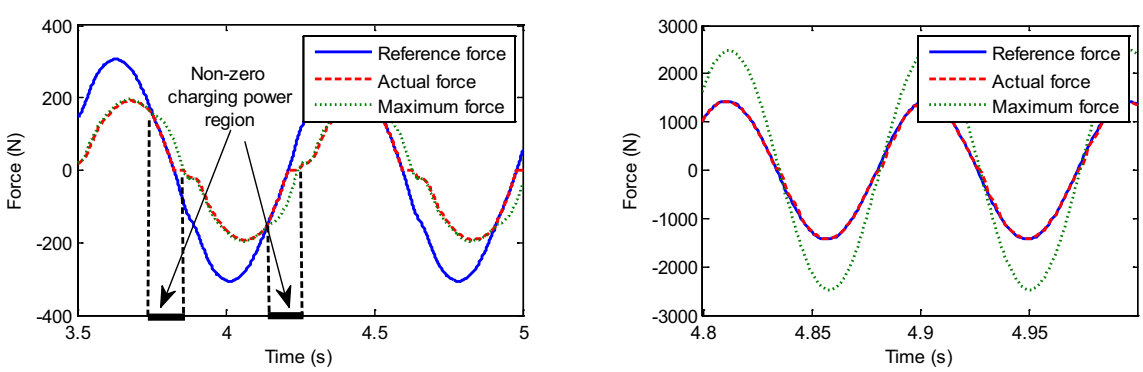
Based on the above discussions in frequency domain, $m = 1.45$ and $n = 0.39$ is selected to ensure the suspension dynamic performance under semi-active control.

Table 2 Simulation parameters

Description	Symbol	Value
Internal resistance	R	10.2 Ω
Winding inductance	L	12.8 mH
Thrust force constant	k_i	77.9 N/A
Back EMF constant	k_e	62.6 V/(m/s)
Nominal voltage of the battery	E_r	12 V
Integral coefficient	c_{bo}, c_{bu}	0.5
Switching gain	k_{bo}, k_{bu}	0.2

Simulation Tests

Simulation tests are conducted to evaluate the control effect. Parameters of the motor and circuit are listed in Table 2.



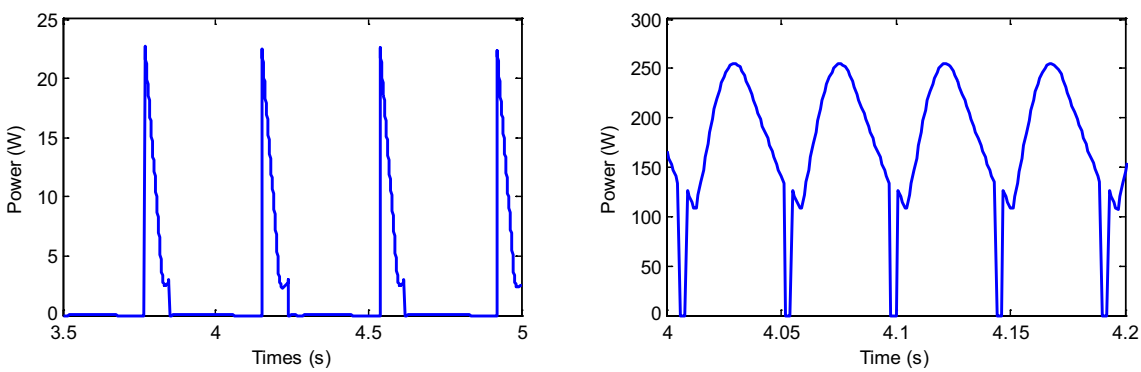
(a) 1.3Hz **(b)** 10.9Hz

Fig. 8 Body acceleration

Sinusoidal Profile Road

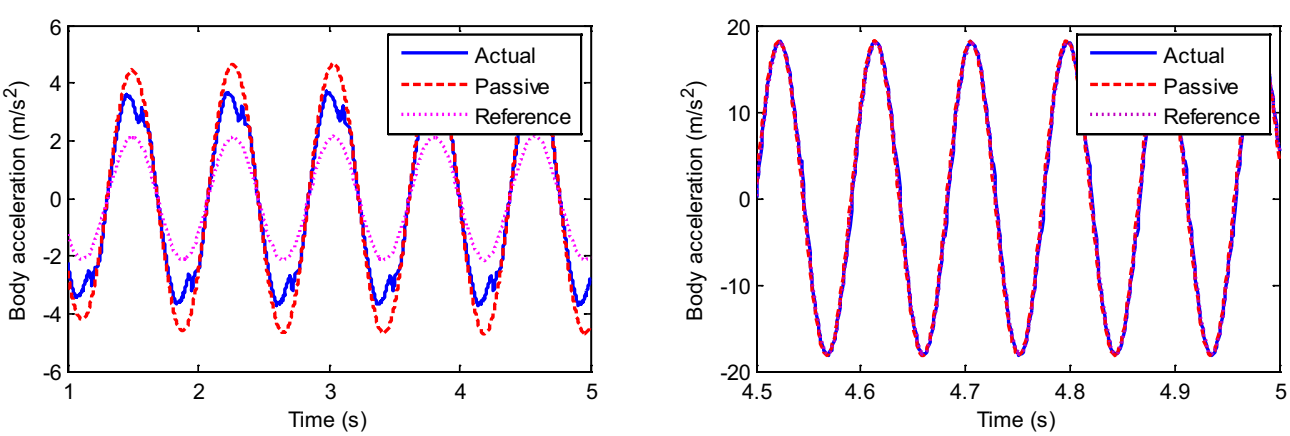
The resonance frequencies of sprung and unsprung masses are 1.3 Hz and 10.9 Hz, respectively. Hence, the 30 mm amplitude sinusoidal road excitations with 1.3 Hz and 10.9 Hz are set as the test input. The results of the proposed system are compared with those of the passive suspension. Figure 8 compares the actual force, the reference force and the maximum electromagnetic damping force. The actual force cannot track the ideal force in most time regions when the excitation frequency is 1.3 Hz, while accurate tracking can be realized at unsprung mass resonance frequency. This situation agrees with the phenomenon of Fig. 7. At 1.3 Hz frequency, α is greater than 1 in most time regions. Semi-active control cannot be realized with the chosen control parameters. The linear motor needs to be powered by the battery to track the reference force. In this case, the motor operation characteristic is represented by $F_{max} - v$ curve. However, there are also time regions that the motor tracks the reference force well. It is because of the obvious phase displacement between the actual force and reference force at 1.3 Hz. Due to the phase displacement, F_{max} is greater than the required control force in these time regions (F_{ref} and F_{max} are in the same direction). $\alpha < 1$ is realized. Therefore, the actual force can track the reference force well.

Figure 9 compares the power that charges the battery under 1.3 Hz and 10.9 Hz sinusoidal excitations. It is found that the charging power is nearly 0 for most of the time at 1.3 Hz. The reason is that at 1.3 Hz, α is greater than 1 and the motor operation characteristic is limited by $F_{max} - v$ curve. It means that both VT1and VT2 switch on and the battery is open circuit. Theoretically, there is no power to charge the battery. However, due to phase displacement during the control as shown in Fig. 8a, the charging power is non-zero in the time regions that $\alpha < 1$. When the excitation frequency is 10.9 Hz, the charging power is larger because of the large back EMF of the motor. It is also found that there are distortion regions where the charging power is 0.



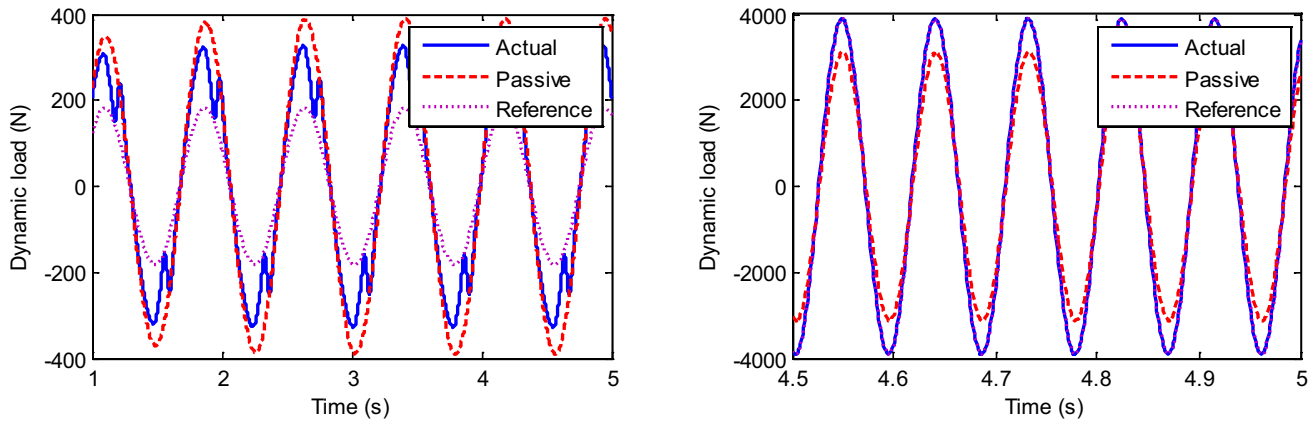
(a) 1.3Hz **(b)** 10.9Hz

Fig. 9 Power charges the battery



(a) 1.3Hz **(b)** 10.9Hz

Fig. 10 Body acceleration



(a) 1.3Hz **(b)** 10.9Hz

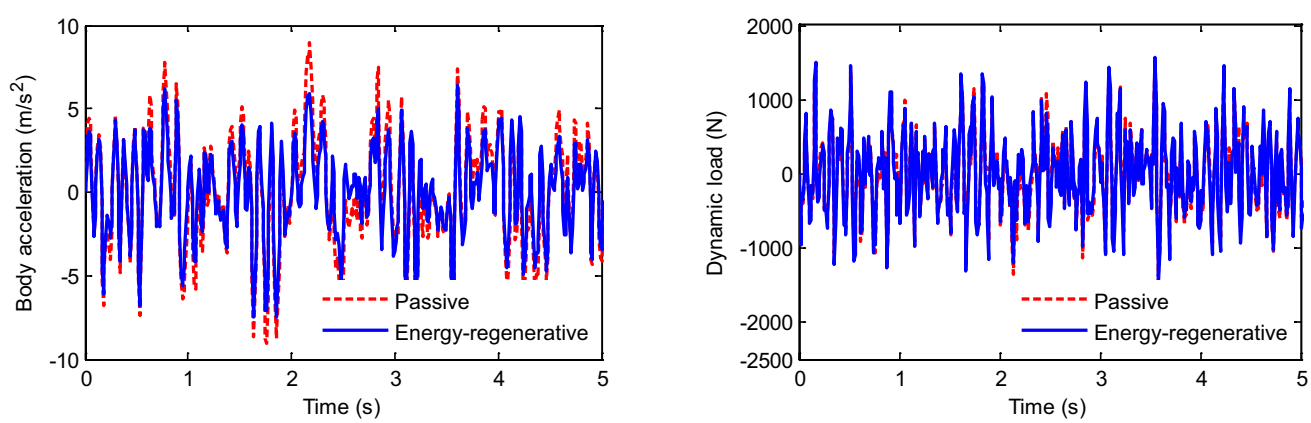
Fig. 11 Tire dynamic load

This distortion is also because of the slight phase displacement. In fact, $\alpha < 1$ happens at 10.9 Hz in nearly all the time regions. From Fig. 9, it is obvious that the proposed suspension has better energy-regenerative performance under high-frequency excitations.

Figures 10 and 11 depict the vehicle body acceleration and tire dynamic load with different frequency excitations. Corresponding to the force tracking situations under the two kind excitations, the actual responses and reference values are inconsistent when the excitation frequency is 1.3 Hz. It is found that the actual body acceleration and tire dynamic load under the proposed semi-active control are all reduced at 1.3 Hz. Both ride comfort and handling performance are improved than the passive suspension. However, the actual control effects are obviously worse than the ideal ones. When the frequency is 10.9 Hz, the actual responses agree with the reference values. At the unsprung mass resonance frequency, although body acceleration is nearly the same as that of passive suspension, the tire dynamic load of the proposed suspension system are apparently increased. It means that handling performance is deteriorated at unsprung mass resonance frequency.

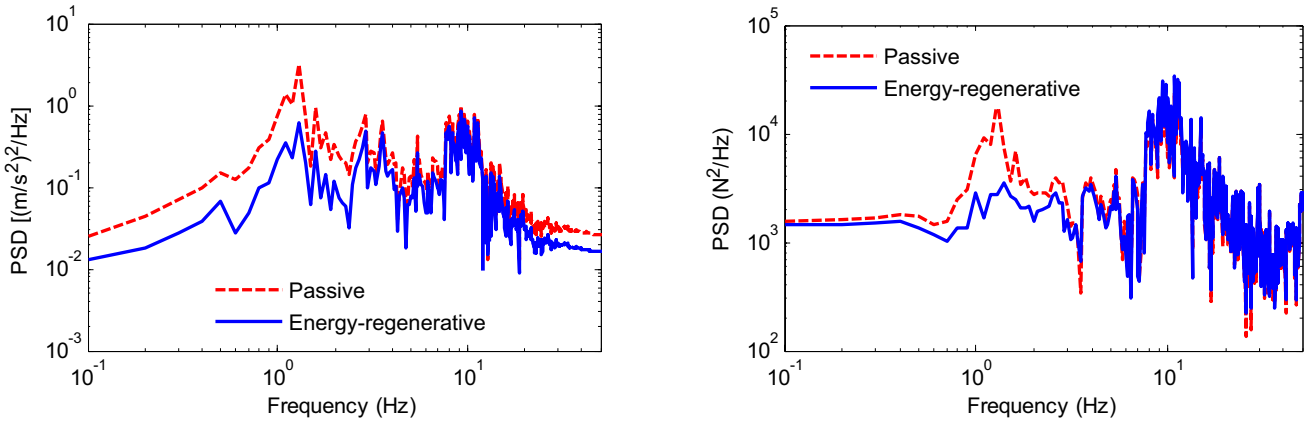
Random Road Input

The scenario that the vehicle drives on D-class road at the speed of 60 km/h is used as test input. Simulation results are shown in Figs. 12 and 13. The body acceleration of energy-regenerative suspension with semi-active control is reduced significantly compared to that of the passive suspension at all frequencies. The tire dynamic load is greatly reduced around sprung mass resonance. By calculating their RMS values, it is found that the body vertical acceleration and tire dynamic load are reduced by 20.32% and −3.9%, respectively. Simulation results of suspension performance agree with the conclusion that there is a trade-off between ride



(a) Body acceleration **(b)** Tire dynamic load

Fig. 12 Responses in time domain



(a) Body acceleration **(b)** Tire dynamic load

Fig. 13 Responses in frequency domain

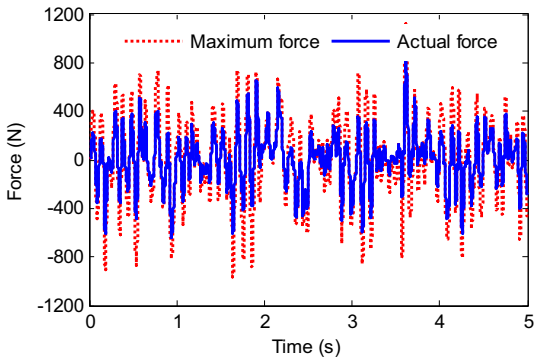


Fig. 14 Actual motor force in comparison with maximum force

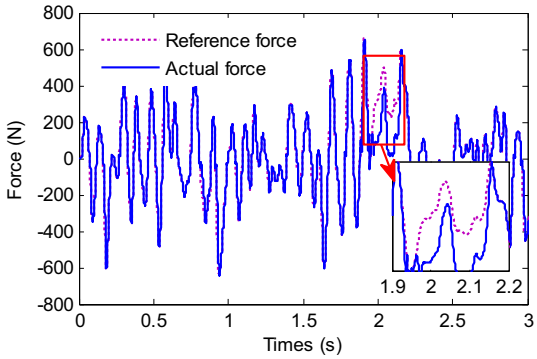


Fig. 15 Actual motor force in comparison with reference force

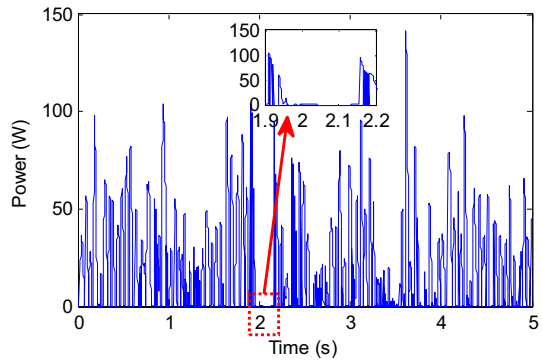


Fig. 16 Power charges the battery

comfort and road-holding performance in suspension control [13]. Although road-holding performance is deteriorated, the deterioration is acceptable.

Figure 14 shows the actual motor force and the maximum force. It is obvious that the actual force to isolate vibration is smaller than the maximum force that the motor could produce as a generator. Therefore, the linear motor can provide enough electromagnetic damping force to isolate the suspension vibration. It is possible for the energy-regenerative suspension to realize semi-active control with proper control parameters in a wide frequency band. The actual electromagnetic damping force also follows the reference force well with the proposed converter and direct current controller, as shown in Fig. 15. In the figure, there are regions where the linear motor cannot track the reference force, which is agreed with the study on the control parameters. For a large actual force, both VTI and VT2 switch on, and the battery is open-circuited. When the required force is in the opposite direction of F_{max}, the actual force is limited to be a very small value.

The power that charges the battery (which is obtained by multiplying the battery voltage by the current flowing through the battery) is shown in Fig. 16. The maximum power can be as much as 141.5 W. There are also regions that the instantaneous power is zero. Figures 14 and 15 indicate that in these regions both VT1 and VT2 switch on

for maximum electromagnetic damping force, or it is the situation that VT1 switches off while VT3 switches on for the opposite direction between maximum electromagnetic damping force and required force. Therefore, the battery is open-circuited and there is no current flowing through the battery in these regions, resulting in the zero charging power. The power that charges the battery is about 20.53 W. Considering the 10 Ω internal resistance, a large amount of energy is consumed by the internal resistance.

Conclusions

Semi-active control is combined with energy-regenerative suspension to improve the suspension dynamic performance. To track the reference force obtained from semi-active control strategy, the working characteristics of the proposed converter in different operation modes are analyzed and the direct current controller is designed. The direct current controller is composed of a mode decision module and a sliding mode module. Analyses of the converter steady characteristics provide theoretical basis for the mode decision strategy. The sliding mode strategy is derived on the basis of the converter average models under CCM.

Semi-active control parameters of the energy-regenerative suspension are selected to increase the effective bandwidth of controllable force by establishing the relations between the semi-active control force and the maximum damping force of the motor. Simulation results under sinusoidal excitation show that the semi-active control cannot be realized when the excitation frequency is very low. However, for high frequencies, the control effect is maintained. On random road surface, the control force can be ideally tracked with the system in nearly all the time history. The suspension with semi-active control improves vehicle ride comfort obviously on random excitation road, while slightly deteriorates road-holding performance. Meanwhile, the energy-regenerative suspension ensures a 20.53 W of energy that charges the battery on D-class road with a 60 km/h velocity.

Acknowledgements This work was supported by the National Natural Science Foundation of China (Grant nos. 51575240, U1664258 and U1564201), the Key Research and Development Program of Jiangsu Province (BE2016149), and the Natural Science Research Funding Project of Jiangsu Higher Education Institutions (17KJD70002).

References

1. Zuo L, Zhang PS (2013) Energy harvesting, ride comfort, and road handling of regenerative vehicle suspensions. J Vib Acoust 135:48–65
2. Shi DH, Pisu P, Chen L, Wang SH, Wang RG (2016) Control design and fuel economy investigation of power split HEV with energy regeneration of suspension. Appl Energy 182:576–589
3. Zhang JQ, Peng ZZ, Zhang L, Zhang Y (2013) A review on energy-regenerative suspension systems for vehicles. Proc World Congr Eng 3:3–5
4. Suda Y, Shiiba T (1996) A new hybrid suspension system with active control and energy regeneration. Veh Syst Dyn 25:641–654
5. Okada Y, Harada H (1996) Regenerative control of active vibration damper and suspension systems. In: Proceedings of 35th IEEE conference on decision and control, vol 4, pp 4715–4720
6. Okada Y, Kim SS, Ozawa K (2003) Energy regenerative and active control suspension. In: ASME 2003 international design engineering technical conferences and computers and information in engineering conference. American Society of Mechanical Engineers, pp 2135–2142
7. Nakano K, Suda Y, Nakadai S (2003) Self-powered active vibration control using a single electric actuator. J Sound Vib 260(2):213–235
8. Nakano K (2004) Combined type self-powered active vibration control of truck cabins. Veh Syst Dyn 41(6):449–473
9. Montazeri-Gh M, Kavianipour O (2014) Investigation of the active electromagnetic suspension system considering hybrid control strategy. Proc IMechE Part C J Mech Eng Sci 228(10):1658–1669
10. Hsieh CY, Bo H, Golnaraghi F, Moallem M (2016) Regenerative skyhook control for an electromechanical suspension system using a switch-mode rectifier. IEEE Trans Veh Technol 65(12):9642–9650
11. Guida D, Pappalardo CM (2015) Control design of an active suspension system for a quarter-car model with hysteresis. J Vib Eng Technol 3(3):277–299
12. Huang Y, Wu X, Na J, Gao GB, Guo Y (2017) Active suspension control for vehicle systems with prescribed performance. J Vib Eng Technol 5(5):407–416
13. Lee S, Kim WJ (2010) Active suspension control with direct-drive tubular linear brushless permanent-magnet motor. IEEE Trans Control Syst Technol 18(4):859–870
14. Davoudi A, Jatskevich J, Rybel TD (2006) Numerical state-space average-value modeling of PWM DC-DC converters operating in DCM and CCM. IEEE Trans Power Electron 21(4):1003–1012
15. Kalaivani R, Lakshmi P, Rajeswari K (2015) An improved type-2 fuzzy logic approach based sliding mode controller for vehicle active suspension system. J Vib Eng Technol 3(4):431–446
16. Wang DW, Gao XF (2015) Liquid tank modelling and attitude controller design of plant protection UAV. J Drain Irrig Mach Eng 33(11):1006–1012
17. Sun LQ, Li ZX, Xu X (2014) Quasi-sliding mode variable structure control and test of semi-active air suspension damping. J Jiangsu Univ (Nat Sci Ed) 35(6):621–626
18. Nguyen SD, Choi SB (2013) A novel minimum–maximum data-clustering algorithm for vibration control of a semi-active vehicle suspension system. Proc IMechE Part D J Automob Eng 227(9):1242–1254
19. Afkar A, Marzbanrad J, Amirirad Y (2015) Modeling and semi active vertical vibration control of a GA optimized 7DoF driver model using an MR damper in a seat suspension system. J Vib Eng Technol 3(1):37–47
20. Chen L, Shi DH, Wang RC, Zhou HW (2016) Energy conservation analysis and control of hybrid active semiactive suspension with three regulating damping levels. Shock Vib 2016:6196542
21. Kawamoto Y, Suda Y, Inoue H, Kondo T (2007) Modeling of electromagnetic damper for automobile suspension. J Syst Des Dyn 1(3):524–535

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.